

Lecture Notes: Counting k -Dimensional Subspaces of an n -Dimensional Vector Space via Group Actions

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1 Introduction

Let V be an n -dimensional vector space over a field K . We want to answer the question:

How many distinct k -dimensional subspaces does V have?

If K is infinite, then for $0 < k < n$ there are infinitely many such subspaces. The interesting counting problem occurs when $K = \mathbf{F}_q$ is a finite field with q elements.

The standard answer is that the number of k -dimensional subspaces of $\mathbf{F}_q^n$ is the Gaussian binomial coefficient

$$\begin{bmatrix} n \\ k \end{bmatrix}_q.$$

In these notes we derive this formula using *group actions*. This viewpoint is conceptually important because it explains the counting formula as an orbit-stabilizer computation for the natural action of $\mathrm{GL}_n(\mathbf{F}_q)$ on the Grassmannian of k -dimensional subspaces.

2 The Grassmannian and the Main Result

Definition 2.1. Let V be an n -dimensional vector space over K . The set of all k -dimensional subspaces of V is called the *Grassmannian* and is denoted by

$$\mathrm{Gr}(k, V).$$

When $V = K^n$, we also write

$$\mathrm{Gr}(k, n) = \mathrm{Gr}(k, K^n).$$

When $K = \mathbf{F}_q$, the set $\mathrm{Gr}(k, n)$ is finite.

Definition 2.2. For $0 \leq k \leq n$, the *Gaussian binomial coefficient* is defined by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1})}.$$

Equivalently,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

Theorem 2.3 (Main Theorem). *Let V be an n -dimensional vector space over $\mathbf{F}_q$. Then the number of distinct k -dimensional subspaces of V is*

$$|\mathrm{Gr}(k, V)| = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Equivalently, if $V \cong \mathbf{F}_q^n$, then

$$|\mathrm{Gr}(k, n)| = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Our goal is to prove this theorem by group actions.

3 Why Group Actions Enter Naturally

The group $\mathrm{GL}_n(\mathbf{F}_q)$ consists of all invertible linear maps

$$\mathbf{F}_q^n \rightarrow \mathbf{F}_q^n.$$

Since invertible linear maps send subspaces to subspaces and preserve dimension, $\mathrm{GL}_n(\mathbf{F}_q)$ acts naturally on $\mathrm{Gr}(k, n)$.

This action is the key idea:

- the set we want to count is $\mathrm{Gr}(k, n)$,

- the acting group is $\text{GL}_n(\mathbf{F}_q)$,
- the action is transitive,
- so $|\text{Gr}(k, n)|$ is an orbit size,
- orbit-stabilizer reduces the problem to computing the stabilizer of one standard k -subspace.

4 Review of Group Actions

Definition 4.1. Let G be a group and X a set. A *left action* of G on X is a map

$$G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x,$$

such that:

- (1) $e \cdot x = x$ for all $x \in X$,
- (2) $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.

Definition 4.2. Let G act on X , and let $x \in X$.

- The *orbit* of x is

$$\text{Orb}(x) = \{g \cdot x : g \in G\}.$$

- The *stabilizer* of x is

$$\text{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

Theorem 4.3 (Orbit-Stabilizer). *If a finite group G acts on a set X , then for every $x \in X$,*

$$|\text{Orb}(x)| = \frac{|G|}{|\text{Stab}(x)|}.$$

Proof. The orbit $\text{Orb}(x)$ is in bijection with the left coset space $G/\text{Stab}(x)$. Indeed,

$$g\text{Stab}(x) \mapsto g \cdot x$$

is well-defined and bijective. Therefore

$$|\text{Orb}(x)| = [G : \text{Stab}(x)] = \frac{|G|}{|\text{Stab}(x)|}.$$

□

5 The Action of $\text{GL}_n(\mathbf{F}_q)$ on $\text{Gr}(k, n)$

Proposition 5.1. *The group $\text{GL}_n(\mathbf{F}_q)$ acts on $\text{Gr}(k, n)$ by*

$$A \cdot W = A(W), \quad A \in \text{GL}_n(\mathbf{F}_q), \quad W \in \text{Gr}(k, n).$$

Proof. If $W \subseteq \mathbf{F}_q^n$ is a k -dimensional subspace and $A \in \text{GL}_n(\mathbf{F}_q)$, then $A(W)$ is again a subspace. Since A is an isomorphism, we have

$$\dim A(W) = \dim W = k.$$

Thus $A(W) \in \text{Gr}(k, n)$. The identity matrix fixes every subspace, and

$$(AB)(W) = A(B(W)),$$

so this is a group action. □

Proposition 5.2. *This action is transitive.*

Proof. Let $W, U \in \text{Gr}(k, n)$. Choose a basis $w_1, \dots, w_k$ of W and extend it to a basis

$$w_1, \dots, w_k, w_{k+1}, \dots, w_n$$

of $\mathbf{F}_q^n$. Similarly choose a basis $u_1, \dots, u_k$ of U and extend it to a basis

$$u_1, \dots, u_k, u_{k+1}, \dots, u_n.$$

There exists a unique linear isomorphism $A \in \text{GL}_n(\mathbf{F}_q)$ such that

$$A(w_i) = u_i \quad (1 \leq i \leq n).$$

Then $A(W) = U$. Hence the action is transitive. $\square$

Remark 5.3. Transitivity means that all k -dimensional subspaces look the same from the point of view of $\text{GL}_n(\mathbf{F}_q)$. So to count all of them, it is enough to understand one convenient representative and its stabilizer.

6 A Standard k -Subspace

Let

$$W_0 = \text{span}(e_1, \dots, e_k) \subseteq \mathbf{F}_q^n,$$

where $e_1, \dots, e_n$ is the standard basis of $\mathbf{F}_q^n$.

Since the action is transitive,

$$\text{Gr}(k, n) = \text{Orb}(W_0).$$

Therefore,

$$|\text{Gr}(k, n)| = |\text{Orb}(W_0)| = \frac{|\text{GL}_n(\mathbf{F}_q)|}{|\text{Stab}(W_0)|}.$$

So the problem reduces to computing $|\text{GL}_n(\mathbf{F}_q)|$ and $|\text{Stab}(W_0)|$.

7 The Size of $\text{GL}_n(\mathbf{F}_q)$

Proposition 7.1. *The order of $\text{GL}_n(\mathbf{F}_q)$ is*

$$|\text{GL}_n(\mathbf{F}_q)| = (q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{n-1}).$$

Proof. An invertible $n \times n$ matrix over $\mathbf{F}_q$ is exactly an ordered basis of $\mathbf{F}_q^n$, written as its column vectors.

To choose such a basis:

- the first column can be any nonzero vector: $q^n - 1$ choices,
- the second column must lie outside the span of the first: $q^n - q$ choices,
- the third column must lie outside the span of the first two: $q^n - q^2$ choices,

and so on. For the i -th column, one must avoid an $(i-1)$ -dimensional subspace containing q^{i-1} vectors, so there are $q^n - q^{i-1}$ choices.

Multiplying gives

$$|\text{GL}_n(\mathbf{F}_q)| = (q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{n-1}).$$

$\square$

8 The Stabilizer of the Standard k -Subspace

We now compute $\text{Stab}(W_0)$, where

$$W_0 = \text{span}(e_1, \dots, e_k).$$

Proposition 8.1. *A matrix $A \in \text{GL}_n(\mathbf{F}_q)$ stabilizes W_0 if and only if it has block form*

$$A = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix},$$

where

$$B \in \text{GL}_k(\mathbf{F}_q), \quad D \in \text{GL}_{n-k}(\mathbf{F}_q), \quad C \in M_{k \times (n-k)}(\mathbf{F}_q).$$

Proof. Write vectors in $\mathbf{F}_q^n$ according to the decomposition

$$\mathbf{F}_q^n = W_0 \oplus W'_0, \quad W'_0 = \text{span}(e_{k+1}, \dots, e_n).$$

Relative to this decomposition, a linear map has block matrix form

$$A = \begin{pmatrix} B & C \\ E & D \end{pmatrix}.$$

Now $A(W_0) = W_0$ if and only if every vector of W_0 is sent into W_0 . This happens exactly when the lower-left block E is zero. Thus

$$A = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}.$$

Such a block upper-triangular matrix is invertible if and only if both diagonal blocks are invertible, that is,

$$B \in \text{GL}_k(\mathbf{F}_q), \quad D \in \text{GL}_{n-k}(\mathbf{F}_q).$$

The block C is arbitrary. □

Corollary 8.2. *The stabilizer has order*

$$|\text{Stab}(W_0)| = |\text{GL}_k(\mathbf{F}_q)| |\text{GL}_{n-k}(\mathbf{F}_q)| q^{k(n-k)}.$$

Proof. The block B can be any element of $\text{GL}_k(\mathbf{F}_q)$, the block D can be any element of $\text{GL}_{n-k}(\mathbf{F}_q)$, and the block C can be any $k \times (n-k)$ matrix. There are

$$q^{k(n-k)}$$

choices for C . Multiplying gives the formula. □

9 Orbit-Stabilizer Computation

We now apply orbit-stabilizer to the action of $\text{GL}_n(\mathbf{F}_q)$ on $\text{Gr}(k, n)$.

Proof of the Main Theorem. Since the action of $\text{GL}_n(\mathbf{F}_q)$ on $\text{Gr}(k, n)$ is transitive,

$$|\text{Gr}(k, n)| = \frac{|\text{GL}_n(\mathbf{F}_q)|}{|\text{Stab}(W_0)|}.$$

Using the formulas already proved,

$$|\text{GL}_n(\mathbf{F}_q)| = \prod_{i=0}^{n-1} (q^n - q^i),$$

and

$$|\text{Stab}(W_0)| = |\text{GL}_k(\mathbf{F}_q)| |\text{GL}_{n-k}(\mathbf{F}_q)| q^{k(n-k)}.$$

Thus

$$|\text{Gr}(k, n)| = \frac{|\text{GL}_n(\mathbf{F}_q)|}{|\text{GL}_k(\mathbf{F}_q)| |\text{GL}_{n-k}(\mathbf{F}_q)| q^{k(n-k)}}.$$

Now substitute

$$\begin{aligned} |\text{GL}_k(\mathbf{F}_q)| &= (q^k - 1)(q^k - q) \cdots (q^k - q^{k-1}), \\ |\text{GL}_{n-k}(\mathbf{F}_q)| &= (q^{n-k} - 1)(q^{n-k} - q) \cdots (q^{n-k} - q^{n-k-1}). \end{aligned}$$

After simplifying, this becomes

$$|\text{Gr}(k, n)| = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}.$$

Therefore

$$|\text{Gr}(k, n)| = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

□

10 A Cleaner Simplification

It is useful to see the simplification more transparently.

Start with

$$|\text{Gr}(k, n)| = \frac{|\text{GL}_n(\mathbf{F}_q)|}{|\text{GL}_k(\mathbf{F}_q)| |\text{GL}_{n-k}(\mathbf{F}_q)| q^{k(n-k)}}.$$

Write

$$\begin{aligned} |\text{GL}_n(\mathbf{F}_q)| &= \prod_{i=0}^{n-1} (q^n - q^i), & |\text{GL}_k(\mathbf{F}_q)| &= \prod_{i=0}^{k-1} (q^k - q^i), \\ |\text{GL}_{n-k}(\mathbf{F}_q)| &= \prod_{i=0}^{n-k-1} (q^{n-k} - q^i). \end{aligned}$$

One may also rewrite

$$|\text{GL}_n(\mathbf{F}_q)| = q^{\frac{n(n-1)}{2}} \prod_{j=1}^n (q^j - 1),$$

and similarly for the smaller general linear groups. Then the powers of q cancel exactly, leaving

$$|\text{Gr}(k, n)| = \frac{\prod_{j=n-k+1}^n (q^j - 1)}{\prod_{j=1}^k (q^j - 1)}.$$

This is another standard form of the Gaussian binomial coefficient:

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1) \cdots (q - 1)}.$$

11 Interpretation of the Stabilizer Formula

The formula

$$|\text{Stab}(W_0)| = |\text{GL}_k(\mathbf{F}_q)| |\text{GL}_{n-k}(\mathbf{F}_q)| q^{k(n-k)}$$

has a clear meaning.

The stabilizer consists of all invertible linear maps that preserve the standard k -subspace W_0 . Such a map may:

- change basis inside W_0 : this contributes a factor $|\text{GL}_k(\mathbf{F}_q)|$,
- change basis inside a complementary $(n - k)$ -space: this contributes $|\text{GL}_{n-k}(\mathbf{F}_q)|$,
- shear the complement into W_0 : this contributes $q^{k(n-k)}$.

So the orbit-stabilizer formula says:

$$\text{number of } k\text{-subspaces} = \frac{\text{all ambient basis changes}}{\text{basis changes preserving one chosen } k\text{-subspace}}.$$

This is the group-action analogue of the more elementary counting argument with ordered bases.

12 Examples

Example 12.1 (Lines in $\mathbf{F}_q^3$). The number of 1-dimensional subspaces of $\mathbf{F}_q^3$ is

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}_q = \frac{q^3 - 1}{q - 1} = q^2 + q + 1.$$

So there are $q^2 + q + 1$ lines through the origin.

Example 12.2 (Planes in $\mathbf{F}_q^3$). The number of 2-dimensional subspaces of $\mathbf{F}_q^3$ is

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}_q = \begin{bmatrix} 3 \\ 1 \end{bmatrix}_q = q^2 + q + 1.$$

This symmetry reflects the fact that k -subspaces and $(n - k)$ -subspaces are counted by the same Gaussian coefficient.

Example 12.3 (Two-dimensional subspaces of $\mathbf{F}_2^4$). For $q = 2$, $n = 4$, $k = 2$,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = \frac{(2^4 - 1)(2^4 - 2)}{(2^2 - 1)(2^2 - 2)} = \frac{15 \cdot 14}{3 \cdot 2} = 35.$$

So $\mathbf{F}_2^4$ has exactly 35 distinct 2-dimensional subspaces.

13 Symmetry

Proposition 13.1. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n - k \end{bmatrix}_q.$$

Proof. This follows directly from the product formulas. It also has a group-theoretic interpretation: the stabilizer of a k -subspace and the stabilizer of a complementary $(n - k)$ -subspace have matching sizes in the orbit calculation. $\square$

14 Connection with the Elementary Counting Proof

There is another classical proof:

- count ordered linearly independent k -tuples in $\mathbf{F}_q^n$,
- divide by the number of ordered bases of a fixed k -space.

That proof gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}.$$

The group-action proof is really the same idea at a more structural level:

- $\mathrm{GL}_n(\mathbf{F}_q)$ encodes all ambient basis changes,
- the stabilizer encodes the changes that preserve one chosen k -subspace,
- orbit-stabilizer turns this into the desired count.

So the group-action method is not only elegant; it explains *why* the quotient formula must appear.

15 Infinite Fields

Proposition 15.1. *If K is infinite and V is an n -dimensional vector space over K , then for $0 < k < n$ the set $\mathrm{Gr}(k, V)$ is infinite.*

Proof. It suffices to consider $k = 1$. In K^2 , each line through the origin is a 1-dimensional subspace. When K is infinite, there are infinitely many slopes, hence infinitely many such lines. Therefore $\mathrm{Gr}(1, 2)$ is infinite, and similarly $\mathrm{Gr}(k, V)$ is infinite whenever $0 < k < n$. $\square$

Remark 15.2. So the finite counting formula is specific to finite fields. Over infinite fields, the Grassmannian is not a finite set but a geometric space.

16 Grassmannians as Homogeneous Spaces

The group-action proof can be summarized in one elegant formula:

$$\mathrm{Gr}(k, n) \cong \mathrm{GL}_n(\mathbf{F}_q) / P_k,$$

where $P_k = \mathrm{Stab}(W_0)$ is the stabilizer of the standard k -subspace. Such a subgroup is called a *parabolic subgroup*.

Thus the Grassmannian is a homogeneous space for $\mathrm{GL}_n(\mathbf{F}_q)$, and its cardinality is

$$|\mathrm{Gr}(k, n)| = \frac{|\mathrm{GL}_n(\mathbf{F}_q)|}{|P_k|}.$$

This is the natural group-theoretic form of the answer.

17 Final Theorem

Theorem 17.1 (Final Form). *Let V be an n -dimensional vector space over $\mathbf{F}_q$. Then the number of distinct k -dimensional subspaces of V is*

$$|\mathrm{Gr}(k, V)| = \begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

Moreover, this count is obtained from the transitive action of $\mathrm{GL}_n(\mathbf{F}_q)$ on $\mathrm{Gr}(k, V)$ by orbit-stabilizer:

$$|\mathrm{Gr}(k, V)| = \frac{|\mathrm{GL}_n(\mathbf{F}_q)|}{|\mathrm{Stab}(W_0)|}.$$

18 Exercises

Exercise 18.1. Prove directly that the action of $\mathrm{GL}_n(\mathbf{F}_q)$ on $\mathrm{Gr}(k, n)$ is transitive.

Exercise 18.2. Let $W_0 = \mathrm{span}(e_1, \dots, e_k)$. Show that

$$\mathrm{Stab}(W_0) = \left\{ \begin{pmatrix} B & C \\ 0 & D \end{pmatrix} : B \in \mathrm{GL}_k(\mathbf{F}_q), D \in \mathrm{GL}_{n-k}(\mathbf{F}_q), C \in M_{k \times (n-k)}(\mathbf{F}_q) \right\}.$$

Exercise 18.3. Compute $|\mathrm{GL}_2(\mathbf{F}_q)|$ and $|\mathrm{GL}_3(\mathbf{F}_q)|$.

Exercise 18.4. Use orbit-stabilizer to compute the number of lines in $\mathbf{F}_q^3$.

Exercise 18.5. Compute $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q$ when $q = 2$ and when $q = 3$.

Exercise 18.6. Prove the symmetry

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q.$$

Exercise 18.7. Explain how the elementary basis-counting proof is related to the orbit-stabilizer proof.

19 Selected Answers

•

$$|\mathrm{GL}_2(\mathbf{F}_q)| = (q^2 - 1)(q^2 - q).$$

•

$$|\mathrm{GL}_3(\mathbf{F}_q)| = (q^3 - 1)(q^3 - q)(q^3 - q^2).$$

•

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q \Big|_{q=2} = \frac{(2^4 - 1)(2^4 - 2)}{(2^2 - 1)(2^2 - 2)} = 35.$$

•

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q \Big|_{q=3} = \frac{(3^4 - 1)(3^4 - 3)}{(3^2 - 1)(3^2 - 3)} = \frac{80 \cdot 78}{8 \cdot 6} = 130.$$

One-Sentence Takeaway

Over a finite field $\mathbf{F}_q$, the k -dimensional subspaces of $\mathbf{F}_q^n$ form a single $\mathrm{GL}_n(\mathbf{F}_q)$ -orbit, and orbit-stabilizer computes their number as the Gaussian binomial coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$.